scientific reports

Check for updates

OPEN

Inhibition of the lateral hypothalamus emboldens adult female spiny mice to huddle with an established group of novel peers

Venezia C. Roshko[1,2], Nicklas A. Gose[1,2] & Aubrey M. Kelly[1]✉

Animals of large group-living species that exhibit dispersal or have overlapping territories with other groups frequently encounter novel conspecifics. To avoid injury, successfully obtain a mate, integrate into a new group, and/or to determine one's social rank, it is crucial to accurately assess social information. The lateral hypothalamus (LH) is critical for learning about food-related cues and shifting behavior toward or away from salient events. While the LH facilitates risk assessment in dyadic social competitions, how the LH modulates social behavior in non-aggressive contexts with novel peers remains unknown. In the highly colonial spiny mouse (*Acomys dimidiatus*), we used chemogenetics to inhibit the LH of females as they interacted with novel peers in a novel vs. familiar preference test, a group size preference test, and a group interaction test. Although control females were investigative and prosocial (e.g., affiliative proximity), they exhibited significant social avoidance of a novel peer group. However, we found that inhibition of the LH induced a preference for social novelty, decreased social avoidance, and promoted affiliative proximity and huddling with a novel, previously established group of peers. These findings suggest that the LH may function to promote cautious behavior, potentially via risk assessment, in novel social environments.

Keywords Social behavior, Spiny mouse, Lateral hypothalamus, Group interaction, Risk assessment

In group-living animals, individuals often disperse and leave the natal home to join a new group[1,2]. Joining a new group, either during juvenility or adulthood (referred to as delayed dispersal), can convey multiple benefits, including inbreeding avoidance, reduced competition, or greater availability of resources[3,4]. Conversely, attempting to join an established group could lead to resource limitation, injury, or even fatality[5,6]. As such, mechanisms that promote social assessment, and perhaps specifically social risk assessment, likely facilitate an animal's ability to successfully join a new group. Yet, little is known about how the brain modulates social assessment, particularly in novel group contexts.

Spiny mice (*Acomys*) are a colonial rodent native to Africa, the Middle East, and southeastern Asia, with group sizes ranging from 12 to 46 in rural desert habitats and 60 + in resource-abundant, urban environments; additionally, spiny mouse groups often exhibit territorial overlap[7–9]. In the lab, spiny mice of the genus *dimidiatus* (formerly referred to as *cahirinus*) exhibit high degrees of sociality and can be housed in mixed-genetic relation, same-sex groups of up to 30 individuals per cage[10]. Spiny mice exhibit low levels of aggression, readily approach novel conspecifics, and are typically prosocial with others regardless of age, novelty/familiarity, kinship, or reproductive/nonreproductive context[11–15]. Further, studies conducted in a lab setting have demonstrated that spiny mouse adults will join previously established groups[16]. Despite the highly affiliative phenotype of this species, spiny mouse sociality does have its limits. For example, huddling in the home cage is typically not observed between two or more novel individuals until after cohabitation for several days (pers. obs.), and acceptance of a newcomer into a previously established group can take up to 1 month[16]. Therefore, although spiny mice are affiliative and socially neophilic, given that they typically do not immediately huddle with novel conspecifics, particularly in group contexts, this suggests that individuals exhibit some level of social discretion, potentially because of risk assessment.

Although the lateral hypothalamus (LH) has been primarily studied in the context of feeding behavior[17,18] several studies have also elucidated a role for the LH in regulating social behavior[19–21]. Studies examining behavior in aggressive contexts have found that the LH promotes violent aggression in cats and rats[22–26], as

[1]Department of Psychology, Emory University, 36 Eagle Row, Atlanta, GA 30322, USA. [2]These authors shared first authorship to this work: Venezia C. Roshko and Nicklas A. Gose. ✉email: aubrey.kelly@emory.edu

well as risk assessment and defensive behavior in rats[27]. Studies have also found that the LH is involved in facilitating social dominance in C57/BL6J mice and may be important for rank learning in social hierarchies[21]. Although in a nonsocial context, previous studies have shown that the LH is critical for learning about food-related cues, and in second order conditioning experiments the LH biases learning toward cues with a greater reward association[28,29]. Given these findings, scientists have argued that the LH is crucial for learning and has the potential to shift behavior toward or away from salient events[30]. Whether the LH modulates *social* learning, however, largely remains unknown, though it is likely that this brain region could play a role in assessment of novel social information.

In the present study, we use chemogenetics (i.e., designer receptors exclusively activated by designer drugs, DREADDs) to examine LH contributions to nonreproductive social behavior in female spiny mice as they interact with novel conspecifics in a group interaction test, a group size preference test, and a novel vs. familiar preference test. We have previously shown that spiny mice exhibit little to no aggression in these contexts[11–15]. Interestingly, little is known about how the LH modulates social behavior in contexts that are largely devoid of aggression; indeed, most studies examining LH-mediated aggression are conducted in high-aggressive strain male rodents. In most mammals, inhibition of the LH decreases aggressive behavior[22,23,31]. However, because spiny mice typically do not exhibit much aggressive behavior during novel interactions, aggression would be at a 'floor' level in these contexts; thus, LH inhibition in spiny mice cannot reduce a behavior that is not being exhibited. Yet, if the LH modulates social assessment, thereby promoting territoriality in aggressive species but perhaps merely cautious behavior in highly social spiny mice, then inhibition of this brain region in spiny mice may result in extreme levels of social boldness – possibly due to a reduction of risk assessment.

We previously found that female spiny mice do not exhibit a preference for novel or familiar same-sex conspecifics[11]. However, because the LH facilitates risk assessment in rats[27] LH activity may promote caution with novel conspecifics and restrict social novelty-seeking behavior in spiny mice. Therefore, we hypothesized that chemogenetic inhibition of the LH would induce a preference for novelty in a novel vs. familiar preference test. Additionally, we hypothesized that LH inhibition would facilitate investigative and affiliative interactions with a novel group of peers during in a group interaction test. Furthermore, spiny mice exhibit a reliable preference to affiliate with large over small groups of novel peers[11,13,15] and we recently found that female spiny mice exhibit greater LH neural responsivity when exposed to a large, compared to a small, peer group[13]. Because the LH may facilitate social assessment, we hypothesized that inhibition of this region would interfere with the inherent preference for female spiny mice to affiliate with large over small groups of peers, thereby eliminating the typical preference of spiny mice to prefer affiliating with a large group in a group size preference test.

Results

To examine contributions of the LH to social behavior in novel social contexts in female spiny mice, subjects were randomly assigned to 1 of 3 conditions: (1) hM4D + CNO – to inhibit the LH (n = 10), (2) hM4D + saline – to control for off-target effects of insertion of an hM4D virus (n = 11), or (3) mCherry + CNO – to control for off-target effects of injection of CNO (n = 10). After 5–8 weeks of viral incubation, females were tested in 3 behavioral assays over 2 days in a randomized order: a novel vs. familiar preference test, a group size preference test, and a novel group interaction. 30 min prior to behavioral testing, subjects received an intraperitoneal (IP) injection of either saline or 10 mg/kg of CNO.

Validation of inhibitory dreadds in spiny mice

We previously validated Cre-dependent DREADDs in spiny mice[13] demonstrating the efficacy of chemogenetic technology in this species. Additionally, we had previously validated the hM4D DREADDs used here in the lateral septum for an unrelated project (unpub. obs.). To confirm that inhibitory hM4D DREADDs sufficiently inhibited neural activity specifically in the LH, 4 subjects from the hM4D + CNO condition and 4 subjects from the mCherry + CNO condition were tested in an immediate early gene (IEG) study after they had completed behavioral testing for the present experiment. We previously found that exposure to a group of 7 novel, same-sex peers elicits a high response of Fos expression in the LH[13]. Thus, for the IEG validation study here, 30 min after a peripheral injection of 10 mg/kg CNO, subjects were placed in a large Plexiglas testing chamber (60.96 × 45.72 × 38.1 cm) with a group of 7 novel, same-sex conspecifics confined under separate wire mesh containers in the center of the chamber. Subjects freely explored the chamber for 90 min and were then immediately perfused. Brain tissue from these subjects was processed via immunohistochemistry to label for the IEG protein Fos, and the percentage of virus-labeled neurons colocalized with Fos in the LH was averaged across 4 tissue sections that were 120 µm apart. A Mann Whitney U-test revealed a significant difference between groups ($Z = -2.09$, $P = 0.021$; Fig. 1A), such that the hM4D condition (Fig. 1B) had a significantly lower percentage of colocalized neurons compared to the control condition (Fig. 1C), confirming the efficacy of the inhibitory DREADDs in the LH of spiny mice. For visualization of viral targeting, see Fig. S1).

Inhibition of the LH induces a preference for social novelty

To test the hypothesis that LH activity may restrict social novelty-seeking in spiny mice, we conducted a novel vs. familiar preference test. For 10 min, females freely explored a large chamber with a novel, same-sex conspecific restrained under a wire mesh container on one side and a familiar, same-sex conspecific under a container on the other side of the chamber. We generated preference scores based on the time spent investigating (paws or nose making contact with the stimulus containers) or affiliating with (time spent resting within 1 body length of stimulus containers) each stimuli. A general linear model (GLM) with condition as a fixed factor yielded a main effect of condition for investigation ($F_{(2,30)} = 3.408$; $P = 0.047$; Fig. 2A); however, after Bonferroni posthoc corrections, there were no significant differences between conditions (all $P > 0.065$). GLM analyses also revealed a main effect of condition for affiliation ($F_{(2,30)} = 4.944$; $P = 0.014$; Fig. 2B). Posthoc analyses showed

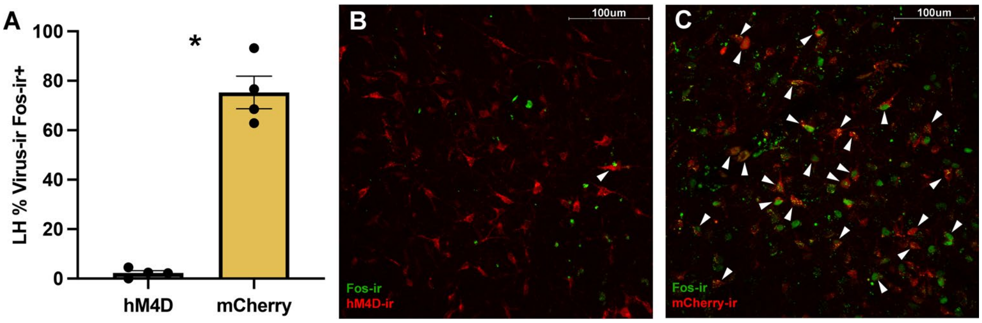

Fig. 1. The hM4D DREADD virus inhibits neural responses to a large group of peers. **(A)** The percentage of virus-immunoreactive (-ir) cells colocalized with Fos in the lateral hypothalamus (LH) after a peripheral injection of CNO in animals that received either hM4D or mCherry virus intracranial injections. **(B)** A representative image of viral and Fos labeling in the LH of a female subject that received the hM4D virus. **(C)** A representative image of viral and Fos labeling in the LH of a female subject that received the mCherry virus. White arrows indicate colocalization of virus and Fos. Data are represented as mean ± SEM. Dots represent individual data points. An asterisk indicates statistical significance of $P \leq 0.05$.

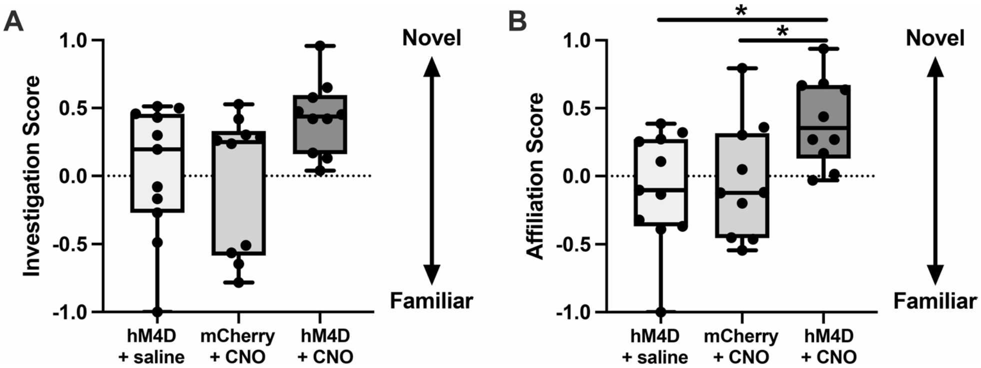

Fig. 2. Inhibition of the LH promotes a preference for affiliation with novel, same-sex conspecifics. **(A)** Investigation scores of female subjects in the hM4D + saline (beige), mCherry + CNO (light green), and hM4D + CNO (dark green) conditions. **(B)** Affiliation scores of female subjects in the hM4D + saline, mCherry + CNO, and hM4D + CNO conditions. Positive scores indicate a propensity for investigating or affiliating with the novel, same-sex conspecific. Data are represented with whiskers indicating the minimum and maximum and solid lines as the median. An asterisk indicates statistical significance of $P \leq 0.05$ for posthoc analyses from the GLM comparing scores across conditions.

that females in the hM4D + CNO (i.e., inhibited LH) condition, exhibited a significantly greater preference for novel, same-sex conspecifics compared to animals in the hM4D + saline control condition ($P = 0.022$) and the mCherry + CNO control condition ($P = 0.050$). To further determine if subjects exhibit novelty or familiarity preferences within conditions, we conducted separate Wilcoxon signed-rank tests for each condition comparing affiliation or investigation with the novel vs. the familiar stimulus animals. Analyses showed no significant preference for affiliation for subjects in the hM4D + saline control condition ($Z = -0.533$, $P = 0.594$) and the mCherry + CNO control condition ($Z = -0.561$, $P = 0.575$) or for investigation for subjects in the hM4D + saline control condition ($Z = -0.178$, $P = 0.859$) and the mCherry + CNO control condition ($Z = -0.663$, $P = 0.508$). However, Wilcoxon signed-rank tests revealed a significant preference for hM4D + CNO animals to affiliate with ($Z = -2.599$, $P = 0.009$) and investigate ($Z = -2.803$, $P = 0.005$) a novel conspecific. These findings demonstrate that inhibition of the LH facilitates a preference to affiliate with a novel peer. Although within-condition analyses demonstrated that LH-inhibited subjects exhibited a preference to investigate novel conspecifics, this preference in the form of an investigation score did not significantly differ from control conditions based on the GLM comparison of investigation scores across conditions, suggesting that inhibition of the LH may not necessarily, or at least not strongly, facilitate investigative preferences.

Subjects were allowed to freely explore the test chamber without interference from any other freely exploring conspecifics. To ensure that the DREADDs did not influence locomotive behavior, we measured the distance traveled and velocity of subjects during the 10 min novel vs. familiar preference test. A GLM found no significant differences between groups for the distance traveled ($F_{(2,30)} = 1.124$; $P = 0.339$) or for the velocity at which an animal traveled ($F_{(2,30)} = 1.098$; $P = 0.347$), indicating that locomotion remained unaltered.

Inhibition of the LH May influence group size preference

To test the hypothesis that the LH promotes social assessment, thereby contributing toward the preference for spiny mice to affiliate with a large group of peers, we conducted a group size preference test. Females freely explored a two-chambered testing apparatus that contained a group of 8 novel, same-sex conspecifics in one chamber and a group of 2 novel, same-sex conspecifics in the other chamber (Fig. 3A). We generated an affiliative preference score based on the time spent affiliating in the social zone of the large group chamber and the social zone of the small group chamber. A GLM with condition (i.e., hM4D + CNO, hM4D + saline, or mCherry + CNO) as a fixed factor yielded no effect of condition on affiliation preference scores ($F_{(2,30)} = 0.828$; $P = 0.447$; Fig. 3B), suggesting that the LH does not contribute toward group size preference in female spiny mice. Additionally, to determine if subjects exhibit group size affiliative preferences within conditions, we conducted separate Wilcoxon signed-rank tests for each condition comparing affiliation with the small vs. the large groups. Analyses yielded a significant preference for the large group in the hM4D + saline control condition ($Z = -2.756$, $P = 0.006$) and the mCherry + CNO control condition ($Z = -2.497$, $P = 0.013$). However, a Wilcoxon signed-rank test failed to show a significant preference for the hM4D + CNO animals ($Z = -0.663$, $P = 0.508$). Although technically not outliers, this effect is likely driven by the 3 subjects that spent substantially more time with the small group. Because the GLM analyses yielded no significant difference in affiliation scores across conditions, we cannot conclusively state whether LH inhibition influences group size preferences; further studies are required to determine the role of the LH in female spiny mouse grouping preferences. Lastly, because we observed that LH inhibition induced a preference for social novelty in the novel vs. familiar preference test, in the group size preference test we also examined the percentage of test time spent in both social zones (both of which were comprised of novel conspecifics), as opposed to time spent not in social zones, to determine if LH inhibition caused females to spend more time in contact with novel conspecifics. However, GLM analyses did not reveal an effect of condition on percentage of social contact time ($F_{(2,30)} = 0.647$; $P = 0.532$).

Inhibition of the LH facilitates affiliation with and decreases avoidance of novel peers

In addition to preference tests, we sought to obtain a rich behavioral dataset characterized by free interactions among individuals in a novel group interaction test, allowing us to further determine the role of the LH in modulating behavior in a novel social context. Subjects were allowed to freely interact with a previously established group of 3 novel, same-sex conspecifics in a large chamber for 1 h. Time sampling of frequencies of distinct types of behavior were obtained every minute, generating 60 datapoints per subject (see Table 1 for an ethogram). We did not run analyses for aggressive behavior because we observed no chasing or fighting for any subject and only 2 instances of subjects being chased.

GLM analyses with condition as a fixed factor yielded no effect of condition for investigation ($F_{(2,30)} = 3.145$; $P = 0.059$; Fig. 4A) but did yield a main effect of condition for subjects being investigated by conspecifics ($F_{(2,30)} = 3.615$; $P = 0.040$; Fig. 4B). Posthoc analyses showed that females in the hM4D + CNO experimental condition were investigated less frequently by novel, same-sex conspecifics compared to animals in the hM4D + saline control condition ($P = 0.038$) but not the mCherry + CNO control condition ($P = 0.861$). Because the experimental inhibition condition did not significantly differ from all control groups, it is difficult to discern whether inhibition of the LH influences behavior in a manner that impacts how that animal is treated by others. Although there were no outliers, the effect observed here may be driven by 3 subjects in the hM4D + saline control condition that experienced a high number of instances in which they were investigated.

We quantified the frequency of events when subjects were active near (i.e., within one body length) a conspecific but were not directly engaging with them. For example, subjects were considered active near another individual if a conspecific was resting and the focal animal was jumping at the chamber wall within one body

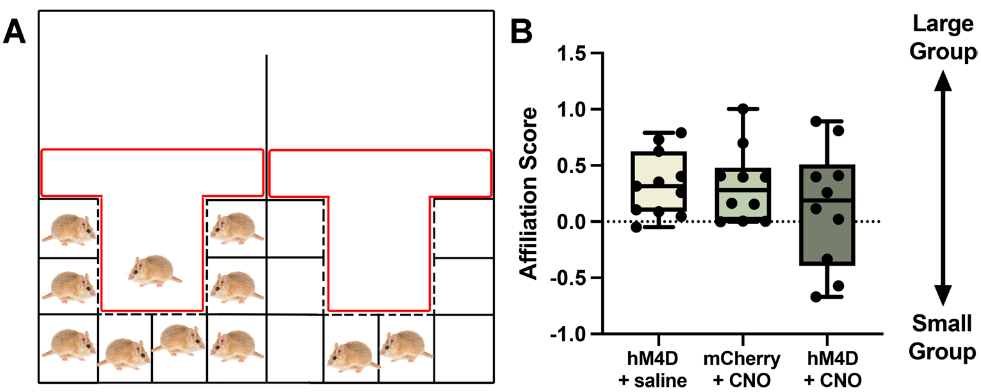

Fig. 3. Inhibition of the LH does not influence group size preference. **(A)** A schematic of the testing apparatus with one chamber containing 8 novel, same-sex conspecifics (left) and the other chamber containing 2 novel, same-sex conspecifics (right). The red areas indicate the social zones used for quantifying time spent with each group. **(B)** Affiliation scores of female subjects in the hM4D + saline (beige), mCherry + CNO (light green), and hM4D + CNO (dark green) conditions. Positive scores indicate a propensity for affiliating with the large group of peers. Data are represented with whiskers indicating the minimum and maximum and solid lines as the median.

Behavior	Description
Fighting	The focal individual lunges or bites a conspecific
Chasing	The focal individual is running after a conspecific rapidly and aggressively whilst the conspecific attempts to flee
Being chased	The focal individual is running away from and attempting to flee a conspecific that is rapidly and aggressively running after them
Following	The focal individual is not making physical contact, but is actively following a conspecific for at least 2 s; the following pace may be walking but not running
Being followed	The focal individual is not making physical contact, but is actively being followed by a conspecific for at least 2 s; the following pace of the conspecific may be walking but not running
Investigation	The focal individual's nose and/or whiskers are in physical contact with a conspecific's body
Being investigated	A conspecific's nose and/or whiskers are in physical contact with the body of the focal individual
Huddling	The focal individual is resting in direct body contact with at least one other conspecific
Avoidance	The focal individual is resting by themselves and is at least half of a chamber length away from other conspecifics. Additionally, the subject relocates to the opposite side of the chamber when conspecifics move toward the space they were resting in alone but the conspecific does not follow the subject
Rest near	The focal individual is resting (immobile for at least 5 s) within one body length of another conspecific but not physically touching. If there was a clear barrier between them (i.e., from the clear acrylic tube), the behavior was still considered as rest near. No specific body orientation of the focal individual toward the conspecific(s) was required to qualify as rest near
Active near	The focal individual is active near another conspecific but not directly engaging with the conspecific. Example 1: two animals feeding in the food tray. Example 2: focal animal jumping at the wall within one body length of a conspecific that is resting.
Grooming	The focal individual is using its mouth or paws to groom any part of the body
Feeding/drinking	The focal individual is consuming food or drinking water.

Table 1. Ethogram for behavioral observations during the group interaction test.

length of the conspecific. A GLM showed no effect of condition on the frequency a subject exhibited active non-overt behavior near another conspecific ($F_{(2,30)} = 1.087$; $P = 0.351$; Fig. 4C). Additionally, we examined subject avoidance of conspecifics in the group, which we operationally defined as subjects being at least half of a chamber-length away from all other conspecifics. Note that the previously established group of 3 individuals primarily remained together, when active and resting, throughout the 1-hour test. Here a GLM revealed a main effect of condition for avoidance ($F_{(2,30)} = 21.335$; $P < 0.001$; Fig. 4D), such that females in the hM4D + CNO experimental condition exhibited fewer avoidance events than females in the hM4D + saline control condition ($P < 0.001$) and the mCherry + CNO control condition ($P < 0.001$). These data show that inhibition of the LH results in less avoidant behavior when females freely interact with a novel group of peers.

Lastly, we examined the consequences of LH inhibition on prosocial behaviors. We quantified the frequency of events when subjects rested within one body length of a conspecific (referred to as "resting near"), as well as huddling, which entails resting in physical contact with at least one other conspecific. GLM analyses yielded a main effect of condition for when subjects rested near other conspecifics ($F_{(2,30)} = 24.851$; $P < 0.001$; Fig. 4E). Female spiny mice in the hM4D + CNO experimental condition rested near novel conspecifics more than females in the hM4D + saline ($P < 0.001$) and the mCherry + CNO ($P < 0.001$) control conditions. Similarly, we found a main effect of condition for huddling ($F_{(2,30)} = 19.900$; $P < 0.001$; Fig. 4F). Again, females in the hM4D + CNO experimental condition huddled with novel conspecifics more than females in the hM4D + saline ($P < 0.001$) and the mCherry + CNO ($P < 0.001$) control conditions. Notably, only a single experimental condition female did not exhibit huddling whereas only 1 of 21 control subjects exhibited any huddling behavior. Together these results demonstrate that LH inhibition facilitates prosocial interactions with novel conspecifics in a peer group.

Discussion

Here we found that inhibition of the LH induced a preference for social novelty, decreased social avoidance of novel peers, and promoted nonreproductive prosocial behavior in female spiny mice. Although unmanipulated spiny mice exhibit prosocial behavior (e.g., affiliative proximity) with novel peers, it typically takes several days for spiny mice to exhibit arguably the most intense form of prosocial behavior - huddling - with a previously established group of same-sex peers. Indeed, only a single control animal in the present study exhibited any huddling with the novel peer group. Yet, females with LH inhibition exhibited relatively high degrees of huddling during the first hour of meeting a novel, previously established group, suggesting that the LH may function to promote cautious behavior, potentially via risk assessment, in novel social environments.

LH involvement in maintaining [social] homeostasis

The hypothalamus is crucial for the regulation of energy homeostasis[32–34]. Although multiple nuclei within the hypothalamus contribute toward meeting physiological homeostatic needs, several studies have demonstrated that the LH is particularly important for the modulation of locomotor activity and energy expenditure[34–36]. Studies with enhanced molecular resolution have further demonstrated that orexin neurons within the LH enable the dynamic modulation of physical activity, as well as feeding[37–39]. Interestingly, LH orexin function is not limited to nonsocial behaviors given that studies have shown that agonism of orexin receptors in the LH increases aggressive behavior whereas orexin receptor antagonism decreases aggressive behavior in CD-1 mice[31]. Further studies have also demonstrated multi-purpose social and nonsocial functions of the LH and showed that subpopulations of leptin receptor and neurotensin neurons balance fulfillment of multiple essential needs, including feeding, drinking, and sociosexual interactions[40]. We likely failed to observe any effect of LH inhibition on feeding/drinking behavior in the present study because our group interaction test was only 1 h in

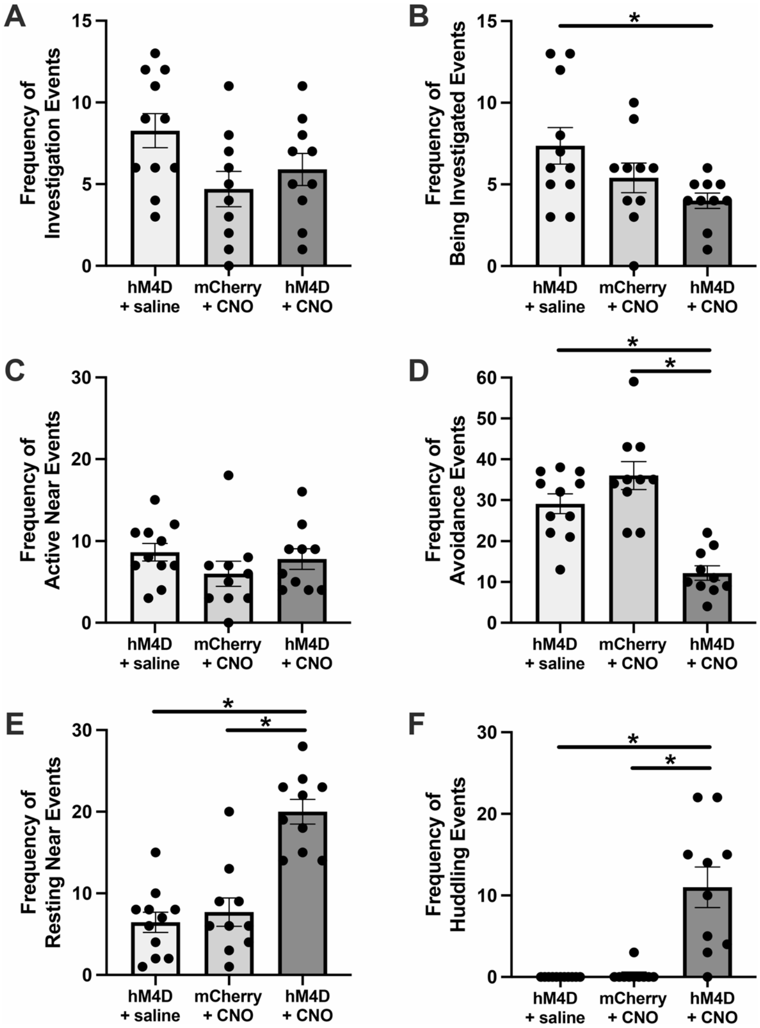

Fig. 4. LH inhibition facilitates prosocial behavior with a novel group. The frequency of events per hour for **(A)** subject-initiated investigation of conspecifics, **(B)** instances when subjects were investigated by conspecifics, **(C)** events when subject were active near, but not directly engaging with, a conspecific, **(D)** instances when a subject avoided all conspecifics in the group, **(E)** events when subjects rested near a conspecifics, and **(F)** instances when subject huddling with at least one other conspecific for females in the hM4D + saline (beige), mCherry + CNO (light green), and hM4D + CNO (dark green) conditions. Data represent instantaneous scan sampling conducted every minute for 1 h. Data are represented as mean ± SEM. Dots represent individual data points. An asterisk indicates statistical significance of $P \le 0.05$ for posthoc analyses from the GLM comparing behavioral frequencies across conditions.

length and our spiny mice subjects, which are a desert-adapted species that can go long periods without food/water, were not food- or water-deprived prior to testing.

Complimentary to the notion that the LH regulates energy homeostasis, researchers have proposed a role for the LH in maintaining social homeostasis, and hypothesize that this brain region may facilitate the maintenance of social connections[19,21] an ability that is especially important in group-living animals with complex hierarchies. Exhibiting social competence allows an animal to appropriately assess a group's social information and to behave in a context-appropriate manner, such as aggressively toward submissive individuals but passively toward dominant peers[41,42]. Studies have revealed nuanced modulation of behavior by the LH, suggesting this brain region is capable of contributing toward the flexible modulation of behavior. In transgenic mice, while LH GABA neurons promote predation, LH glutamate neurons promote predictive evasion of predators[43]. Similarly, GABAergic LH pathways promote real-time place preference, whereas glutamatergic LH pathways facilitate real-time place avoidance in transgenic mice[44]. Unfortunately, we lack cellular resolution in the present study and thus are unable to disentangle cell type- and circuit- specific functions of the LH in spiny mice. However, these prior studies demonstrate that the LH is a dynamic brain region capable of promoting opposing behaviors and could play an important role in promoting the behavioral flexibility required to appropriately interact with group members of varying social rankings[21,42].

Consistent with the hypothesis that the LH is maintaining homeostatic needs and/or driving animals to reach homeostatic setpoints[19,21] the LH may contribute towards the facilitation of behavior that is most appropriate for a given context. For example, in a territorial species, the most appropriate behavior to exhibit toward an intruder coming onto one's territory is to aggressively displace the intruder. Indeed, LH Fos expression positively correlates with attacks toward an intruder in resident rats[25]. Meanwhile, when the focal rat is the intruder, the LH facilitates risk assessment and defensive behaviors[27]. However, in a highly colonial species, if an individual is dispersing and attempting to join a new group, particularly one that is already established, the most appropriate behavior would presumably be to exhibit caution and maintain some distance from potential new group members while the individual gathers social information. Indeed, it can take up to 1 month for an adult spiny mouse newcomer to become fully integrated into a new group[16]. Consistent with this, here we observed cautious behavior in control female spiny mice during the group interaction test (see Fig. 4); as a newcomer, although subjects engaged in investigative and prosocial interactions, on average the behavior they exhibited the most frequently was avoidance. Interactions among conspecifics are highest during the first ~ 30 min of a 1-hr long group interaction test, followed by a tendency to find a place to rest, with the occasional bout of exploration or seeking of food/water. In control females, after the initial investigation phase, most spiny mice settled into an inactive period of rest, with the subjects choosing to rest a substantial distance away from the group. However, in females with an inhibited LH, significantly less socially avoidant behavior was observed and instead females exhibited unusually socially bold behavior and joined the group huddle when resting. Together, this growing body of literature suggests that the LH may not directly modulate pro- or anti- social behaviors but rather may play a role in facilitating context-appropriate social behavior.

LH involvement in assessment

Because the LH regulates learning about food-related cues[28,29] it has been posited that this brain region is crucial for learning and directing behavior toward salient objects or events[30,44]. Extending this hypothesis to the social realm, studies have shown that the LH encodes 'self' and 'other' information in primates[20] and promotes social preferences in rats[45]. Pharmacological inactivation of the LH abolishes novel vs. familiar social preferences in male Sprague Dawley rats[45] suggesting that LH inactivation may have eliminated the ability for the rats to assess or learn social information, thereby preventing them from being able to make appropriate social decisions or ultimately exhibit social preferences. Further, when rats are intruders in a resident-intruder paradigm, lesioning the LH specifically reduces risk assessment, but does not impact social investigation[27]. In the present study, we also failed to find significant effects of LH inhibition on investigative behavior in both the novel vs. familiar preference test and the group interaction test but did identify effects of LH inhibition resulting in increased social affiliation. Although investigation may be inherently intertwined with assessment, assessment is the evaluation of information that has been gathered via investigation, and these related processes may be differentially regulated in the brain. If LH inhibition causes reduced assessment, these subjects may not be evaluating information gathered from investigations. Notably, the study that found that LH inactivation eliminated risk assessment in rats defined risk assessment as exhibiting crouch-sniffs, in which a rat is immobile with an arched back and actively sniffing the environment, and stretch-postures, where the rat slowly stretches the body toward the stimulus[27]. In the present study, we did not specifically score risk assessment, nor did we quantify behaviors such as crouch-sniffs or stretch-postures. Spiny mice have a body shape that is surprisingly dissimilar to lab mice and rats; the neck region of a spiny mouse is quite compact in most adults, and we do not reliably observe behaviors akin to a stretch that is quantifiable to the naked eye. Instead, spiny mice move their whole bodies, usually quite quickly, toward stimuli of interest, which is why we scored behaviors such as following (quickly keeping pace behind a conspecific, but not making contact) or investigation (i.e., making bodily contact with the nose). Thus, future studies, ideally using computer vision to identify subtle changes in postural dynamics, are required to specifically measure risk assessment behaviors in spiny mice. Here we suggest that our findings may indicate that the LH is broadly involved in social assessment – the evaluation of another conspecific – which does not necessarily need to be accompanied by stretching postures. Further, because we did not test our subjects in a classic social recognition test, such as a habituation-dishabituation test, we cannot conclude whether LH inhibition interfered with social recognition abilities. However, given that manipulation of this brain region induced a preference to affiliate with novel conspecifics, this suggests that spiny mice were indeed able to distinguish between conspecific types. Thus, our results contribute to the growing body of literature that suggests that the LH is important for learning and plays a role in the social assessment of novel conspecifics.

In preference tests, spiny mice typically run from stimulus to stimulus, evenly distributing investigation across stimuli for the first few minutes of the test; if they exhibit a preference, they then rest next to the preferred stimulus animal for the remainder of the test. We previously found that spiny mice do not exhibit a preference for novel or familiar same-sex conspecifics[11]. Here, however, although females with LH inhibition did not investigate stimulus animals more or less than control animals did, they reliably chose to affiliate more with the novel conspecific in the preference test. If the LH regulates assessment, inhibiting this brain region may have reduced risk assessment, in turn also reducing cautious behavior, and thus resulting in an increase in social boldness. In a preference test in which stimulus animals are restrained under mesh containers, it is unclear why a female spiny mouse with likely disrupted assessment abilities would prefer a novel over a familiar same-sex conspecific. However, further insight may be gained from considering the free interactions between conspecifics in the group interaction test.

In an experimental setting where it is not an option to leave a behavioral test, the ultimate goal of being introduced to a new group is likely to join and be accepted into that group because spiny mice are much more likely to survive in a group than on their own, as is the case for most species that evolved to live in large groups[46]. Presumably, the safest way to join a new group is to first gather information and then to exhibit caution and potentially even distant or submissive behaviors[2]. Indeed, we observed some prosocial and investigative behaviors in control females, but these animals mostly exhibited avoidant behavior and maintained distance from the group. Similar to the novel vs. familiar preference test, we observed no differences in investigation in control and experimental females in the group interaction test. However, again we saw that females with LH inhibition exhibited high degrees of social boldness, which manifested as little social avoidance and huddling.

Together, along with previous findings that lesioning the LH did not influence social investigation in rats[27] these findings suggests that the LH does not directly regulate investigative behavior but instead may direct an animal to appropriately use information gathered from investigation. Thus, while control and experimental animals did not significantly differ in investigation as measured in our study, their behavior was different after social information was gathered (i.e., they did or did not exhibit a social preference or huddled more/less), supporting the hypothesis that the LH may facilitate risk assessment. Interestingly, as it turns out, there may not have been a significant amount of risk involved in interacting with a novel group of peers given that the peer groups readily huddled with the LH-inhibited newcomer females. Although we did not test our subjects in a classic test to assess anxiety-like behavior (e.g., open field or elevated plus maze), our results suggest that a delay to join a new peer group is primarily due to cautious or social anxiety-related behavior on behalf of the newcomer and not due to aggression or territoriality exhibited by the group. Indeed, females with LH inhibition successfully huddled with the previously established group demonstrating high levels of tolerance from the group. Notably, a same-sex peer group in this species may entail little competition, as studies have shown that same-sex groups in spiny mice do not exhibit stable social hierarchies[47] unlike lab mice, which exhibit linear social hierarchies[48]. However, it is possible that group tolerance levels would differ significantly in mixed-sex groups given that the pregnancy status of females influences the rate at which newcomer males are accepted into established groups[16].

If the LH facilitates assessment, particularly of novel conspecifics, then we may not have observed an effect of LH inhibition on behavior in the group size preference test because all stimuli in the test were novel. Notably, there was more variation in the preference scores of females in the experimental group, suggesting that LH inhibition may have interfered with social preferences. Indeed, analyses within conditions yielded no group size preference for LH-inhibited females but did show the typical preference for the large group in animals in the two control conditions. However, because the overall model comparing affiliation scores across conditions was not significant, we cannot definitively conclude that manipulation of the LH altered group size preferences. A larger sample size is Likely required to confirm that inhibition of the LH eliminates the preference for female spiny mice to affiliate with large over small groups. We previously found that female spiny mice exhibit greater Fos expression in the LH when exposed to a group of 7 novel peers compared to a single novel peer, suggesting that this brain region may be involved in the robust preference for spiny mice to prefer affiliating with larger groups[13]. However, the results from the current study suggest that greater LH Fos responses to a group may reflect the fact that there are simply more animals to evaluate in a group of 7 peers, and brain regions up- and/or down- stream of the LH may be responsible for directly promoting affiliative preferences.

This study bolsters the notion that the LH is much more than a hub for modulating defensive behaviors and feeding[30]. Rather, the LH is a brain region capable of processing multiple types of stimuli and may play an especially crucial role in assessment and learning, allowing animals to respond to dynamic environments in a context-appropriate manner. The findings presented here support the hypothesis that the LH is critical for learning and expand upon the literature demonstrating LH-regulation of food-related learning by providing data that suggests an additional role for the LH in social assessment.

Materials and methods

Animals

37 adult female spiny mice (*Acomys dimidiatus*) age post-natal day (PND) 325–425 were used as subjects; ages were distributed evenly across conditions. 6 subjects were removed from the dataset due to surgical mistargeting. Animals were obtained from our breeding colony, which originates from breeders obtained from the captive-bred colony of Dr. Ashley W. Seifert (University of Kentucky). All subjects were housed in pairs in standard rat polycarbonate cage (40.64 × 20.32 × 20.32 cm) Lined with Sani-Chips bedding, and were provided with nesting material, rodent igloos, and shepherd shacks, as well as ad Libitum food and water. The animal colony rooms were maintained on a 14-h Light: 10-h dark cycle with an ambient temperature of 25 ± 2 °C. The circadian rhythm of *Acomys* species varies depending on environmental conditions[49,50]. In our colony, spiny mice exhibit diurnal activity patterns and were tested during the day under the light phase. All procedures were approved by the Institutional Animal Care and Use Committee of Emory University. All methods were conducted in accordance with relevant ARRIVE guidelines and regulations. Only females were used for this study due to limited availability of males in our breeding colony at the time experiments were conducted. Thus, future studies are required to determine how the LH modulates social behavior in male spiny mice.

Experimental design

Female spiny mice were randomly assigned to 1 of 3 groups: (1) hM4D + CNO – to inhibit the LH ($n = 10$), (2) hM4D + saline – to control for off-target effects of insertion of an hM4D virus ($n = 11$), or (3) mCherry + CNO – to control for off-target effects of injection of CNO ($n = 10$). Females underwent stereotaxic surgery and were allowed 5 weeks to recover and for the virus to incubate. Within 8 weeks of surgery, subjects underwent behavioral testing in 3 assays over 2 days in a randomized order: a group size preference test, a novel vs. familiar preference test, and a novel group interaction. 30 min prior to behavioral testing, subjects received an intraperitoneal (IP) injection (100ul fluid/10 g body weight) of either saline or 10 mg/kg of CNO (dissolved in DMSO and diluted in sterile saline). All testing was completed within 2 h after injection. To avoid multiple injections in a single testing day, the 1-hour novel group interaction test was completed on one day 30 min after injections. On a second testing day, 30 min after injections, animals were tested in the 10-min group size preference and 10-min novel vs. familiar preference tests, with a 20–30 min break in-between; the order of the preference tests was randomized across subjects. Additionally, the order of testing days was randomized across subjects. After all testing was completed, subjects were perfused and brains collected for histological confirmation of surgical targeting. All methods were performed in accordance with the relevant guidelines and regulations.

Stereotaxic injections

The hM4D adeno-associated virus (AAV8-pAAV-hSyn-hM4D(Gi)-mCherry; titre 2.2E[13] GC/mL; a gift from Bryan Roth (Addgene plasmid #50475)) or mCherry virus (AAV8-pAAV-hSyn-mCherry) were intracranially injected at 4 target sites (bilaterally at different rostral/caudal positions) in volumes of 200nL into the LH. Rostral coordinates: −0.69 mm anterior, 1.3 mm lateral, depth − 5.3 mm. Caudal coordinates: −1.27 mm anterior, 1.3 mm lateral, depth − 5.6 mm. For all intracranial injections, after anesthetization via inhalation of 4% isoflurane, subjects received a subcutaneous injection of 1.5 mg/kg dose of Alloxate™ between the shoulders as an analgesic. Anesthetization was confirmed via observation of a slow and steady breathing rate and no response to toe pinching[51]. Subjects were fitted to a stereotax and maintained under anesthesia at 2% isoflurane. AAVs were delivered through a pulled glass pipette at a rate of 1nl/sec using a Nanosyringe (Drummond Nanoject III). Pipettes were held at the injection site for 10 min following AAV release. Coordinates were slightly adjusted based on mouse size and viral vectors were allowed to express at 5–8 weeks prior to behavioral testing. All surgeries were completed within 90 min.

Histology, immunohistochemistry, and neural quantification

After behavioral testing, subjects were chemically euthanized via isoflurane overdose and Transcardially perfused with chilled 0.1 M phosphate buffer saline (PBS) followed by chilled 4% paraformaldehyde. Brains were post-fixed in 4% paraformaldehyde overnight, incubated in 30% sucrose for 48 h at 4 °C, and were then frozen in Tissue-Tek O.C.T. and stored at −80 °C prior to sectioning. For targeting confirmation, brains were sectioned at 40 μm and mounted to subbed slides for subsequent visualization of hM4D or mCherry fluorescence. For the 8 animals that also underwent the validation IEG study, brains were sectioned at 40 μm and processed via immunohistochemistry as previously described[12,15]. Briefly, after tissue was rinsed in 0.1 M PBS it was incubated in a blocking solution (10% normal donkey serum + 0.03% Triton X-100 in PBS) for 1 h at room temperature. Tissue was then incubated for 48 h at 4 °C in a primary antibody solution (1:500 rabbit c-Fos, Synaptic Systems + 5% normal donkey serum + 0.03% Triton X-100 in PBS). After primary antibody incubation, tissue was rinsed in PBS and then incubated in a secondary antibody solution (1:250 donkey anti-rabbit conjugated to Alexa Fluor 488, Thermo Fisher + 5% normal donkey serum + 0.03% Triton X-100 in PBS) for 2 h at room temperature in the dark. After rinses in PBS, tissue was mounted on subbed slides with Prolong Gold Antifade with DAPI. Photomicrographs were taken using a Zeiss AxioImager II microscope fitted with an apotome. Four 10X images were taken of the LH where hM4D and mCherry virus was expressed. The percentage of virus-labeled cells (red) colocalized with Fos (green) was quantified and averaged across all 4 sections.

Novel vs. familiar preference test

Subjects were transferred via a plastic beaker into the center of a large Plexiglas testing chamber (60.96 × 45.72 × 38.1 cm) that had, along the longest direction of the chamber, a novel, same-sex conspecific under a wire mesh container on one side and a familiar, same-sex conspecific (i.e., their cagemate) under a wire mesh container on the opposite side. The subject was allowed to freely explore for 10 min. Behavior was video-recorded from a top-down view and the amount of time spent within 1 body length (affiliation) of each stimulus animal and the time spent investigating the stimulus animals (i.e., paws or nose making contact with the stimulus container) was quantified using Behavioral Observation Research Interactive Software (BORIS)[52]. To ensure that the DREADDs did not influence locomotive behavior, we measured distance traveled and velocity using Ethovision XT 17 (Noldus, USA). All tests were scored by a single observer, blinded to treatment.

Group size preference test

We used a custom group size preference chamber that was previously validated in male and female spiny mice[13]. The chamber consists of an acrylic start box (15.24 × 15.24 × 45.72 cm) that releases the subject into a large, opaque acrylic chamber (55.88 × 71.12 × 45.72 cm) divided into two identical sections. Each section contains walls of 8 individual cubicles to hold spiny mice (or other objects). These cubicles are made of clear acrylic with 0.12 cm diameter holes to allow the subject to have restricted access to the stimulus animals. One section contained 8 novel, same-sex conspecifics whereas the other section contained only 2 novel, same-sex conspecifics; the 2 novel, same-sex conspecifics were always placed in the same chambers so that the section containing the small group was identical for all subjects. Subjects were placed into the start box prior to release into the chamber, and were then allowed to freely explore for 10 min. Behavior was video-recorded from a top-down view and the amount of time spent in the social zone of each chamber (e.g., large and small; see red zones in Fig. 3A) was quantified using Ethovision XT 17 (Noldus, USA). All tests were scored by a single observer, blinded to treatment.

Novel group interaction test

Stimulus groups of 3 adult same-sex conspecifics were established by co-housing 2 siblings and 1 unrelated animal for at least 1 month prior to testing. The stimulus group was acclimated for 20 min to a large Plexiglas testing chamber (60.96 × 45.72 × 38.1 cm) that contained a clear acrylic shelter, food, and a water bottle. Subjects were then Transferred via a plastic beaker into the testing chamber and allowed to freely interact with the group for 1 h. 6 stimulus groups were used throughout the experiment and were counterbalanced across subject condition. Behavior was video-recorded from a top-down view and BORIS was used to quantify behavior. Behavior was quantified by time sampling; the video was paused every minute and the behavior exhibited by the subject was recorded. This yielded 60 data points per subject for subject-initiated behavior. In addition to these 60 subject-initiated time samples, we also recorded whether stimulus animals were directing behavior toward the subject (i.e., investigating, following, or chasing the subject). Therefore, a subject was always recorded as exhibiting a behavior, but some time samples also included a stimulus-initiated behavior directed toward the subject. For example, a subject could have been investigating one stimulus animal while simultaneously being investigated by another stimulus animal. See Table 1 for an ethogram. For analyses, we combined "following" and "investigation" for a total measure of investigation; similarly, we combined "being followed" and "being investigated" for a total measure of being investigated. We observed no grooming, fighting, chasing, feeding/drinking, and only 2 instances of being chased, so these behaviors were not statistically analyzed. Note that animals were not food- or water- deprived and that spiny mice are adapted to desert-living and typically do not feed/drink during behavioral assays that are 1 h or less. All tests were scored by a single observer, blinded to treatment.

Statistical analysis

For the novel vs. familiar preference test and the group size preference test, normalized preferences scores were calculated. For example, a normalized preference score for the novel vs. familiar test was calculated as: (Time investigating novel conspecific − time investigating familiar conspecific)/(Time investigating novel conspecific + time investigating familiar conspecific). Normalized scores from these tests were analyzed using GLMs with condition as a fixed factor, and pairwise comparisons were adjusted using the Bonferroni correction. To examine preferences within conditions (i.e., did a specific treatment condition exhibit a preference for novel or familiar conspecifics), we conducted Wilcoxon signed-rank tests. For the group interaction test, raw frequency data were analyzed using GLMs with condition as a fixed factor; all posthoc pairwise comparisons were adjusted using the Bonferroni correction. We initially ran GLMs with age as a fixed factor but observed no significant influence of age on any measure and therefore dropped age as a factor from our models to increase statistical power. We screened the data for outliers, defined as 3 standard deviations outside the mean, however, no outliers were identified. Data were analyzed using SPSS 29 (IBM Analytics, USA) and figures made using Prism 10 (GraphPad, USA).

Data availability

Data are available upon request from the corresponding author.

Received: 10 June 2025; Accepted: 12 September 2025
Published online: 16 October 2025

References

1. Ebensperger, L. A. & Hayes, L. D. On the dynamics of rodent social groups. *Behav. Processes*. **79**, 85–92 (2008).
2. Gaines, M. S. & McClenaghan, L. R. Dispersal in small mammals. *Annu. Rev. Ecol. Evol. Syst.* **11**, 163–196 (1980).
3. Emlen, S. T. The evolution of helping.1. An ecological constraints model. *Am. Nat.* **119**, 29–39 (1982).
4. Kingma, S. A. Food, friends or family: what drives delayed dispersal in group-living animals? *J. Anim. Ecol.* **87**, 1205–1208 (2018).
5. Maag, N. et al. Dispersal decreases survival but increases reproductive opportunities for subordinates in a cooperative breeder. *Am. Nat.* **199**, 679–690 (2022).
6. Ramakrishnan, A. P. & Dispersal-Migration *Encyclopedia Ecol. (Second Edition)* **1**, 185–191 (2008).
7. Shargal, E., Kronfeld-Schor, N. & Dayan, T. Population biology and Spatial relationships of coexisting spiny mice (*Acomys*) in Israel. *J. Mammal.* **81**, 1046–1052 (2000).
8. Shkolnik, A. Studies in the comparative biology of Israeli's two species of spiny mice (genus Acomys). *Ph.D. dissertation, The Hebrew University, Jerusalem, Israel (in Hebrew, English summary)* (1966).
9. Shkolnik, A. & Borut, A. Temperature and water relations in 2 species of spiny mice (*Acomys*). *J. Mammal*. **50**, 245–255 (1969).
10. Haughton, C. L., Gawriluk, T. R. & Seifert, A. W. The biology and husbandry of the African spiny mouse (*Acomys cahirinus*) and the research uses of a laboratory colony. *J. Am. Assoc. Lab. Anim. Sci.* **55**, 9–17 (2016).
11. Fricker, B. A., Seifert, A. W. & Kelly, A. M. Characterization of social behavior in the spiny mouse, *Acomys Cahirinus*. *Ethology* **00**, 1–15 (2021).
12. Fricker, B. A., Ho, D., Seifert, A. W. & Kelly, A. M. Biased brain and behavioral responses towards kin in males of a communally breeding species. *Sci. Rep-Uk*. **13**, 17040 (2023).
13. Fricker, B. A., Murugan, M., Seifert, A. W. & Kelly, A. M. Cingulate to septal circuitry facilitates the preference to affiliate with large peer groups. *Curr. Biol.* **34**, 4452–4463 (2024).
14. Fricker, B. A., Boender, A. J., Young, L. J. & Kelly, A. M. Not just for bonding: nucleus accumbens Oxytocin receptors facilitate huddling with strangers and feeding in male spiny mice. *Psychoneuroendocrinol* **178**, 107496 (2025).
15. Gonzalez Abreu, J. A. et al. Species-typical group size differentially influences social reward neural circuitry during nonreproductive social interactions. *iScience* **25**, 104230 (2022).
16. Cizkova, B., Sumbera, R. & Frynta, D. A new member or an intruder: how do Sinai spiny mouse (Acomys dimidiatus) families respond to a male newcomer? *Behaviour* **148**, 889–908 (2011).
17. Qualls-Creekmore, E. & Munzberg, H. Modulation of feeding and associated behaviors by lateral hypothalamic circuits. *Endocrinology* **159**, 3631–3642 (2018).
18. Stuber, G. D. & Wise, R. A. Lateral hypothalamic circuits for feeding and reward. *Nat. Neurosci.* **19**, 198–205 (2016).
19. Matthews, G. A. & Tye, K. M. Neural mechanisms of social homeostasis. *Ann. N Y Acad. Sci.* **1457**, 5–25 (2019).
20. Noritake, A., Ninomiya, T. & Isoda, M. Representation of distinct reward variables for self and other in primate lateral hypothalamus. *Proc. Natl. Acad. Sci. U S A*. **117**, 5516–5524 (2020).
21. Padilla-Coreano, N. et al. Cortical ensembles orchestrate social competition through hypothalamic outputs. *Nature* **603**, 667–671 (2022).
22. Biro, L. et al. Task division within the prefrontal cortex: distinct neuron populations selectively control different aspects of aggressive behavior via the hypothalamus. *J. Neurosci.* **38**, 4065–4075 (2018).

23. Haller, J. The role of the lateral hypothalamus in violent intraspecific Aggression-The glucocorticoid deficit hypothesis. *Front. Syst. Neurosci.* **12**, 26 (2018).
24. Haller, J. The neurobiology of abnormal manifestations of aggression–a review of hypothalamic mechanisms in cats, rodents, and humans. *Brain Res. Bull.* **93**, 97–109 (2013).
25. Tulogdi, A. et al. Brain mechanisms involved in predatory aggression are activated in a laboratory model of violent intra-specific aggression. *Eur. J. Neurosci.* **32**, 1744–1753 (2010).
26. Tulogdi, A. et al. Neural mechanisms of predatory aggression in rats-implications for abnormal intraspecific aggression. *Behav. Brain Res.* **283**, 108–115 (2015).
27. Rangel, M. J. Jr., Baldo, M. V., Canteras, N. S. & Hahn, J. D. Evidence of a role for the lateral hypothalamic area Juxtadorsomedial region (LHAjd) in defensive behaviors associated with social defeat. *Front. Syst. Neurosci.* **10**, 92 (2016).
28. Sharpe, M. J. et al. Lateral hypothalamic GABAergic neurons encode reward predictions that are relayed to the ventral tegmental area to regulate learning. *Curr. Biol.* **27**, 2089–2100e5 (2017).
29. Sharpe, M. J., Batchelor, H. M., Mueller, L. E., Gardner, M. P. H. & Schoenbaum, G. Past experience shapes the neural circuits recruited for future learning. *Nat. Neurosci.* **24**, 391–400 (2021).
30. Sharpe, M. J. The cognitive (lateral) hypothalamus. *Trends Cogn. Sci.* **28**, 18–29 (2024).
31. Bai, F. et al. Prelimbic area to lateral hypothalamus circuit drives social aggression. *iScience* **26**, 107718 (2023).
32. MortonG.J. Hypothalamic leptin regulation of energy homeostasis and glucose metabolism. *J. Physiol.* **583**, 437–443 (2007).
33. Roh, E., Song, D. K. & Kim, M. S. Emerging role of the brain in the homeostatic regulation of energy and glucose metabolism. *Exp. Mol. Med.* **48**, e216 (2016).
34. Tran, L. T. et al. Hypothalamic control of energy expenditure and thermogenesis. *Exp. Mol. Med.* **54**, 358–369 (2022).
35. Qualls-Creekmore, E. et al. Galanin-Expressing GABA neurons in the lateral hypothalamus modulate food reward and noncompulsive locomotion. *J. Neurosci.* **37**, 6053–6065 (2017).
36. Zink, A. N., Bunney, P. E., Holm, A. A., Billington, C. J. & Kotz, C. M. Neuromodulation of orexin neurons reduces diet-induced adiposity. *Int. J. Obes. (Lond).* **42**, 737–745 (2018).
37. Kotz, C. M., Teske, J. A., Levine, J. A. & Wang, C. Feeding and activity induced by orexin A in the lateral hypothalamus in rats. *Regul. Pept.* **104**, 27–32 (2002).
38. Kotz, C. M. Integration of feeding and spontaneous physical activity: role for orexin. *Physiol. Behav.* **88**, 294–301 (2006).
39. Kotz, C. M. et al. Orexin A mediation of time spent moving in rats: neural mechanisms. *Neuroscience* **142**, 29–36 (2006).
40. Petzold, A., van den Munkhof, H. E., Figge-Schlensok, R. & Korotkova, T. Complementary lateral hypothalamic populations resist hunger pressure to balance nutritional and social needs. *Cell. Metab.* **35**, 456–471e6 (2023).
41. Lee, W. et al. Effect of relative social rank within a social hierarchy on neural activation in response to familiar or unfamiliar social signals. *Sci. Rep.* **11**, 2864 (2021).
42. Zheng, H. et al. Behavioral tests for the assessment of social hierarchy in mice. *Front Behav. Neurosci.* **19**, 1549666 (2025).
43. Li, Y. et al. Hypothalamic circuits for predation and evasion. *Neuron* **97**, 911–924e5 (2018).
44. Nieh, E. H. et al. Inhibitory input from the lateral hypothalamus to the ventral tegmental area disinhibits dopamine neurons and promotes behavioral activation. *Neuron* **90**, 1286–1298 (2016).
45. Barretto-de-Souza, L. et al. Melanin-concentrating hormone and orexin shape social affective behavior via action in the insular cortex of rat. *Psychopharmacol. (Berl).* **242**, 929–943 (2025).
46. Salguero-Gomez, R. More social species live longer, have longer generation times and longer reproductive windows. *Philos. Trans. R Soc. Lond. B Biol. Sci.* **379**, 20220459 (2024).
47. Varholick, J. A. et al. Social dominance status and social stability in spiny mice (Acomys cahirinus) and its relation to ear-hole regeneration and glucocorticoids. *BioRxiv* https://doi.org/10.1101/2022.09.13.507818 (2022).
48. Williamson, C. M., Lee, W. & Curley, J. P. Temporal dynamics of social hierarchy formation and maintenance in male mice. *Anim. Behav.* **115**, 259–272 (2016).
49. Cohen, R. & Smale, L. Kronfeld-Schor, N. Plasticity of circadian activity and body temperature rhythms in golden spiny mice. *Chronobiol Int.* **26**, 430–446 (2009).
50. Levy, O., Dayan, T. & Kronfeld-Schor, N. The relationship between the golden spiny mouse circadian system and its diurnal activity: an experimental field enclosures and laboratory study. *Chronobiol Int.* **24**, 599–613 (2007).
51. Sotocinal, S. G. et al. The rat grimace scale: a partially automated method for quantifying pain in the laboratory rat via facial expressions. *Mol. Pain.* **7**, 55 (2011).
52. Friard, O. & Gamba, M. B. O. R. I. S. A free versatile open-source event-logging software for video/audio coding and live observations. *Methods Ecol. Evol.* **7**, 1325–1330 (2016).

Acknowledgements

This work was supported by the National Science Foundation (IOS-2310626 to AMK).

Author contributions

V.C.R.: conceptualization, data curation, methodology, investigation, writing – reviewing and editing; N.A.G.: data curation, methodology, investigation, writing – reviewing and editing; A.M.K.: conceptualization, investigation, formal analysis, project administration, funding acquisition, supervision, writing – original draft.

Funding

This work was supported by the National Science Foundation (IOS-2310626 to AMK).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-025-20283-4.

Correspondence and requests for materials should be addressed to A.M.K.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.